

Data communication: components, signals and modes ★ Golden

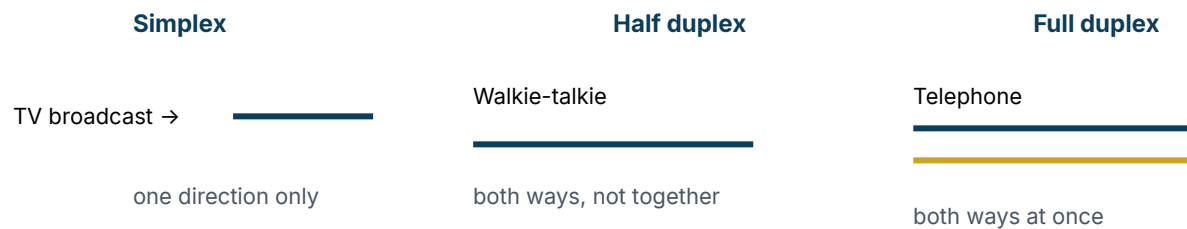
One college period · 40–45 minutes · Computer Communication & Networks

What you will learn

- Define data communication and its five components
- Differentiate analog and digital signals
- Explain simplex, half duplex and full duplex
- Define modulation, demodulation, synchronous and asynchronous transmission

Data communication. The exchange of data between two devices through a transmission medium according to a protocol.

Five components (board list): Sender, Receiver, Message, Medium, Protocol.



Direction of data flow.

Mode	Direction	Example
Simplex	One way only	Radio, keyboard → CPU, TV broadcast
Half duplex	Both ways, not at the same time	Walkie-talkie
Full duplex	Both ways at the same time	Telephone, mobile call

Analog vs digital

Analog signal	Digital signal
Continuous wave, infinite values	Discrete 0 and 1 pulses
Human voice, old telephone line	Computers, optical fibre (light pulses)
Noise distorts easily	More reliable, regenerable

Modulation. Converting a digital signal into analog so it can travel on a telephone line (ASK/FSK/PSK). Performed by a modem on send.

Demodulation. Converting the analog signal back to digital at the receiving modem.

MODEM = MODulator + DEModulator.

Asynchronous	Synchronous
Start and stop bits around each character	A timing clock; data sent in blocks/frames
Cheaper, slower, keyboard typing	Faster, used on networks

Broadcast: one sender, many receivers (TV, radio). Point-to-point: one sender, one receiver (phone call).

HOW BIEK ASKS THIS

2026 shorts: analog vs digital; modulation/demodulation; broadcast vs point-to-point; components of data communication; half duplex MCQ.

COMMON BOARD MISTAKE

Full duplex is both directions simultaneously. Half duplex is both directions but one at a time. Students mix these every year.

اردو میں سمجھیں

ڈیٹا کمیونیکیشن میں بھیجنے والا، وصول کرنے والا، پیغام، میڈیم اور پروٹوکول ہوتے ہیں۔

Simplex ایک طرف، Half duplex باری باری دونوں طرف، Full duplex ایک ساتھ دونوں طرف۔ Modulation ڈیجیٹل کو اینالاگ بناتا ہے، Demodulation الٹ۔

CHECK YOURSELF

1. Telephone is which mode?
2. Modem expands to?

Answers: 1. Full duplex · 2. Modulator Demodulator

MCQs

1. A mode that allows two directions but not simultaneously is:
 - a. Full duplex
 - b. Half duplex
 - c. Synchronous
 - d. Simplex
2. Converting analog to digital is called:
 - a. Modulation
 - b. Telecommunicating
 - c. Demodulation
 - d. Switching
3. Protocol means:
 - a. A topology
 - b. A set of rules
 - c. A router
 - d. A modem

Key: 1=b, 2=c, 3=b

Short questions (3 marks each)

1. Name the components of data communication.
2. Differentiate analog and digital signals.
3. What is modulation and demodulation?
4. Differentiate broadcast and point-to-point.
5. Differentiate synchronous and asynchronous transmission.

Model short answers

1. Sender, receiver, message, medium, protocol.
2. Analog is continuous; digital is 0/1. Digital resists noise better.
3. Modulation: digital→analog for the line. Demodulation: analog→digital. Modem does both.
4. Broadcast: one-to-many. Point-to-point: one-to-one.
5. Async uses start/stop bits per character. Sync uses a clock and sends blocks — faster.

Long questions (board Section C)

1. Explain data communication. Describe communication modes and analog/digital signals with diagrams.